

Alitas Express Full-Stack Ordering Website

Project Overview

Alitas Express is a full-stack restaurant ordering platform designed for a local wings delivery business. The goal of the application is to provide customers with a simple and convenient way to browse available wing flavors, place delivery orders, and contact the business through a modern web interface.

The platform replaces traditional flyer-based marketing with a digital solution that improves customer accessibility and order management.

Business Problem

The business currently relies on promotional flyers and phone orders. This approach makes it difficult to:

- Track customer orders
- Manage order history
- Scale the business
- Provide customers with an online ordering experience
- Collect customer information for future marketing

The website solves these problems by digitizing the ordering process.

System Goals

The system should:

1. Display restaurant information.
2. Showcase available wing flavors.
3. Allow customers to place delivery orders online.
4. Store customers and order information in a database.
5. Provide backend APIs for order processing.
6. Support future expansion such as payments and order tracking.

Functional Requirements

Customer Features

View Menu

Customers can browse available wing flavors:

- Classic
- BBQ
- Buffalo
- Hot
- Lemon Pepper
- Mango Habanero
- Teriyaki

Place Orders

Customers can submit:

- Name
- Phone Number
- Delivery Address
- Flavor Selection
- Quantity

Contact Business

Customers can:

- Call the restaurant
- View business hours
- View delivery information

Non-Functional Requirements

- Responsive Design
- Mobile Friendly
- Secure API Endpoints
- Fast Load Times
- User-Friendly Interface
- Maintainable Code Structure

System Architecture

Client (Frontend)



REST API



Express Server



PostgreSQL Database

Frontend

Responsibilities:

- Display menu items
- Collect customer information
- Submit orders
- Show success/error messages

Technologies:

- React
- Next.js
- Tailwind CSS

Backend

Responsibilities:

- Receive order requests
- Validate user input
- Store orders
- Return API responses

Technologies:

- Node.js
- Express.js

Database

Responsibilities:

- Store customer data
- Store order records

- Store order status

Technology:

- PostgreSQL

Database Design

Orders Table

Field	Type
id	Integer
customer_name	VARCHAR
phone_number	VARCHAR
address	TEXT
flavor	VARCHAR
quantity	Integer
order_status	VARCHAR
created_at	TIMESTAMP

API Design

Create Order

POST /api/orders

Request:

```
{  
  "name": "Tania Correa",  
  "phone": "555-123-4567",  
  "address": "123 Main St",  
  "flavor": "BBQ",  
}
```

```
"quantity": 12
}
```

Response:

```
{
  "message": "Order created successfully"
}
```

Get Orders

GET /api/orders

Returns all customer orders.

Get Order By ID

GET /api/orders/:id

Returns a specific order.

User Workflow

1. Customer visits website.
2. Customer views menu.
3. Customer selects flavor and quantity.
4. Customer enters delivery information.
5. Frontend sends request to backend API.
6. Backend validates data.
7. Order is saved to PostgreSQL.
8. Customer receives confirmation message.

Future Improvements

- Online payments
- Stripe integration
- Order tracking
- Customer accounts
- Admin dashboard
- Inventory management
- SMS notifications
- Analytics dashboard

Learning Objectives

This project demonstrates:

- Full-Stack Development
- REST API Design
- Database Design
- Frontend Development
- Backend Development
- Client-Server Communication
- Software Architecture Principles
- Web Application Deployment